8 ☆

8) $r(t)$, $\dot{r}_t$ $\longrightarrow v = r_t \omega$

$$r_{net}^2 = \dot{r}_t^2 + r_t^2 \omega^2$$

$$\frac{\partial \mathcal{L}}{\partial \dot{r}_t}\dot{r}_t - \mathcal{L} = \mathcal{E}$$

$$\sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\dot{q}_i - \mathcal{L} = \mathcal{E} \implies q_i = r_t$$

$x = r\theta$
$v = r\omega$

$$\mathcal{L} = \frac{1}{2}m(\sqrt{r_{net}})^2$$

$$\mathcal{L} = \frac{1}{2}m((\dot{r}_t)^2 + (r_t\omega)^2)$$

$$\mathcal{E} = \frac{\partial \mathcal{L}}{\partial \dot{r}_t}\dot{r}_t - \mathcal{L}$$

$$\mathcal{E} = m\dot{r}_t(\dot{r}_t) - \frac{1}{2}m[\dot{r}_t^2 + (r_t\omega)^2]$$

$$= m\dot{r}_t^2 - \frac{1}{2}m\dot{r}_t^2 - \frac{1}{2}m r_t^2\omega^2$$

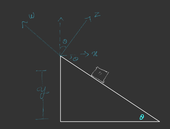
$$\boxed{\mathcal{E} = \frac{1}{2}m\dot{r}_t^2 - \frac{1}{2}m r_t^2\omega^2}$$

Here, $\mathcal{E}$ is not the energy as the system isn't closed. ∵ the rotating stick is being externally driven at const. ω.

Mistakes: 1) Adding non-needed (M, L)

2) Using v_i instead of $\dot{r}_i$ (confusion, mixed up)

7 ☆☆



$$\mathcal{L} = T - V - V(z)$$

$$= \frac{1}{2}m(\dot{\omega}^2 + \dot{z}^2) - mg(\omega\sin(\theta) + z\cos(\theta)) - V(z)$$

As: $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{\partial \mathcal{L}}{\partial q_i}$

i) $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\omega}} = \frac{\partial \mathcal{L}}{\partial \omega}$

$$\frac{d}{dt}(m\dot{\omega}) = -mg\sin(\theta)$$

$$\implies m\ddot{\omega} = -mg\sin(\theta) \quad -(i)$$

ii) $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{z}} = \frac{\partial \mathcal{L}}{\partial z}$

$$\implies m\ddot{z} = -mg\cos(\theta) - \frac{dV}{dz}$$

$$\implies m\ddot{z} = -mg\cos(\theta) - \frac{dV}{dz} \quad -(ii)$$

constraint $\equiv z = 0$

$$\implies z = \ddot{z} = 0$$

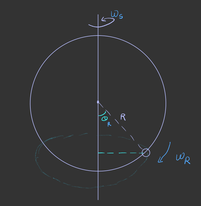
∴ As eq (ii) $\implies m\ddot{z}^{\,\to 0} = -mg\cos(\theta) - \frac{dV}{dz}$

$$\implies \boxed{F_z = mg\cos(\theta)} \quad \left[\because F_z = -\frac{dV}{dz}\right]$$

Hence Proved!

11 ☆☆

Incomplete !!



during equil. $\dot{\theta}_R = 0$

$\theta_R = \theta$

Top down view

$\theta_R = \omega_R t$

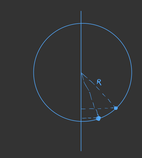
$\theta_s = \omega_s t$

$$T = \frac{1}{2}I_s\omega_s^2 + \frac{1}{2}I_R\omega_R^2$$

$$T = \frac{1}{2}(I_s\omega_s^2 + I_R\omega_R^2)$$

$$V = -mgR\cos(\theta_R)$$

if center of mass = origin



$$\mathcal{L} = T - V$$

$$\mathcal{L} = \frac{1}{2}(I_s\omega_s^2 + I_R\omega_R^2) + mgR\cos(\theta_R)$$

$$\mathcal{L} = \frac{1}{2}(mR^2\sin^2(\theta_s)\omega_s^2 + mR^2\omega_R^2) + mgR\cos(\theta_R)$$

$$\mathcal{L} = \frac{1}{2}mR^2(\sin^2(\theta_s)\dot{\theta}_s^2 + \dot{\theta}_R^2) + mgR\cos(\theta_R)$$

$$\mathcal{L} = \frac{1}{2}mR^2(\sin^2(\theta)\omega^2 + \dot{\theta}^2) + mgR\cos(\theta)$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{\partial \mathcal{L}}{\partial \theta}$$

$$\cancel{m}R^2\ddot{\theta} = \cancel{m}R^2\sin(\theta)\cos(\theta)\omega^2 - \cancel{m}gR\sin(\theta)$$

$$\boxed{R\ddot{\theta} = \sin(\theta)[R\cos(\theta)\omega^2 - g]}$$

At equil: $\dot{\theta} = 0 = \ddot{\theta}$

$\cos(\theta) = \frac{g}{R\omega^2}$ or $\sin(\theta) = 0$

i.e. $\theta = 0; \pi$

i.e. $0 \le \frac{g}{R\omega^2} \le 1$

$\implies \omega^2 \ge \frac{g}{R}$ (or) $\theta = 0; \pi$

Mistakes:

→ notation confusion

→ $a \times b = 0$
∴ $a = 0$
or
$b = 0$

→ Potential sign